

BSBINS401 - Analyse and Present Research Information

Session 5: Advanced Exploratory Data Analysis & Report Preparation

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Learning Objectives

1. Explore advanced approaches to Exploratory Data Analysis (EDA) in Python.
2. Learn techniques to refine and interpret analysis results.
3. Begin structuring the Assessment 1 report, incorporating a clear data narrative.
4. Develop familiarity with correlation analysis and group-based insights.

Assessment 1 Reminder

Assessment 1 Due: Week 7

Components:

1. A report detailing your data cleaning, analysis steps, and findings
2. An in-class presentation summarizing your key insights

Session Overview

1. Common pitfalls in data analysis and how to avoid them
2. Advanced Pandas features (groupby, pivot tables)
3. Correlation and basic hypothesis testing concepts
4. Structuring the final report
5. Hands-on notebook walkthroughs and discussions

1. Common Pitfalls in Data Analysis

- **Overlooking Data Types:**
 - Ensure columns are correctly typed (numeric vs. categorical vs. datetime)
 - Convert string-based numbers (e.g., "25") to int
- **Assuming Missing Data is Random:**
 - Investigate patterns—missingness might be systematic
- **Failing to Document Changes:**
 - Maintain a clear record of each cleaning, filtering, or transformation step
- **Relying on a Single Visualization:**
 - Use multiple plot types to confirm or challenge your initial assumptions

2. Advanced Pandas Tools

Groupby and Pivot Tables

- **groupby()**
Perform split-apply-combine operations (e.g., average by department)
- **pivot_table()**
Reshape data and compute aggregated statistics in a spreadsheet-like pivot format

Example: Grouping Employee Data by Department

```
import pandas as pd

df = pd.read_csv("employee_attrition.csv")
dept_stats = df.groupby("Department")["MonthlyIncome"].mean()

print("Average Monthly Income by Department:")
print(dept_stats)
```

Applying Pivot Tables for Deeper Insights

```
pivot_example = pd.pivot_table(  
    df,  
    values="MonthlyIncome",  
    index="EducationField",  
    columns="Gender",  
    aggfunc="mean"  
)  
  
print("Pivot Table Example:\n", pivot_example)
```

- Quick way to compare numeric outcomes across multiple dimensions (e.g., gender + education)
- Helps spot potential trends or disparities

3. Correlation Analysis

Why Correlation?

- Measures the linear relationship between pairs of variables
- Helps identify features that might be important in your research questions

Creating a Correlation Matrix

```
import seaborn as sns
import matplotlib.pyplot as plt

corr_matrix = df.corr()
plt.figure(figsize=(10, 8))
sns.heatmap(corr_matrix, annot=True, cmap="Blues")
plt.title("Correlation Matrix of Dataset")
plt.show()
```

- **Interpretation:**
 - Values range from -1 to +1
 - Closer to +1 indicates strong positive correlation, -1 indicates strong negative correlation
 - 0 means no linear relationship

4. Structuring Your Assessment 1 Report

Section	Components
1. Introduction	• Research questions/objectives • Overview of dataset
2. Methodology	• Data cleaning steps and justification • Tools and libraries used
3. Analysis & Findings	• EDA techniques, visualizations, correlation insights • Observations and potential explanations
4. Conclusions & Recommendations	• Summarize key insights • Suggest actions or further research
5. References & Appendices	• Citation of data sources and external references

5. Hands-On Notebook Walkthrough

1. Data Wrangling

- Convert columns to correct data types
- Handle outliers or erroneous values

2. Exploratory Plots

- Bar plots of grouped means, standard deviations
- Correlation heatmap for feature relationships

3. Interpreting Results

- Identify top drivers or patterns
- Tie outcomes back to your research questions

Activity: Building Your Draft Analysis

1. Group Analysis:

- Using your dataset, apply a groupby or pivot table to examine at least one important metric by group (e.g., average exam score by major, average income by department).

2. Correlation Matrix:

- Generate a correlation heatmap to see how variables relate. Spot any strong correlations or interesting negative relationships.

3. Discussion:

- Share any surprising findings or confirm previously known insights.

Key Takeaways

- **Advanced EDA:**
 - groupby, pivot tables, correlation
- **Data Narrative:**
 - Connect your plots and stats back to research questions
- **Report Preparation:**
 - Start drafting your methodology and initial findings now

Additional Resources

Resource	Description	Link
Pandas Documentation	Detailed guide on groupby, pivot tables	groupby
Kaggle Learn	Advanced data cleaning & pivot tables	Kaggle: Pandas
Seaborn Docs	Correlation heatmaps and advanced plots	Seaborn Examples

Questions & Next Steps

- **Any Questions?**
 - Email me at jordan.hill@nmtafe.wa.edu.au
 - Office Hours: Mon-Fri, 9 AM - 5 PM
- **Next Steps:**
 - Continue refining your EDA.
 - Incorporate new findings into your Assessment 1 draft.
 - Prepare to share partial analyses or rough visualizations for feedback in the next session.